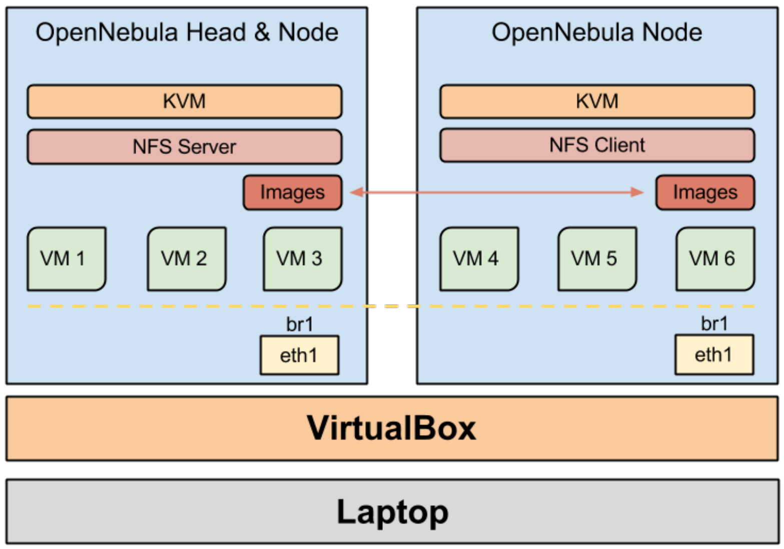
TP – First contact with a Cloud Platform
ING3 - ICC – Cloud Infrastructures
Academic year 2024–2025

Virtual Lab



OpenNebula's administration environment

Installing OpenNebula's sandbox

- Download and start the *OpenNebula VirtualBox Sandbox*:
    - VirtualBox image in `https://s3.amazonaws.com/opennebula-sandbox/OpenNebula-5.2-Sandbox-VBOX.ova`
    - Import the ova image into VirtualBox and boot it.
    - You can download the OpenNebula 5.2 operation guide from this link: `https://docs.opennebula.io/pdf/5.2/opennebula_5.2_operation_guide.pdf`.
- In a browser in your local machine, open OpenNebula's administration interface, called **Sunstone**: http://localhost:9869
    - **login:** `oneadmin`
    - **pasword:** `opennebula`

- Although it is recommended to create a user account, we will keep working with the admin account, only for testing purposes.

Discovering Sunstone

1 Deploy two VMs, **vm1** et **vm2** with the default parameters:

- Template `ttylinux`.
- Capacity 0.1cpu, 256MB memory.
- Hard drive 40 MB.
- `Cloud` network interface.

Open the VMs console. Try to connect from one VM to the other one (ssh, ping) to explain their network configuration:

- Interfaces and networks of each VM.
- Interfaces and networks of the frontend (sandbox VirtualBox) (**login:** `root`, **password:** `opennebula`)
- Connection between the three of them.

□

Basic administration with Sunstone

2

- Open an administrator session.
- Create a Virtual Network called `external` in the address range `192.168.1.0`.
- Start a new VM exclusively with the `external` interface attached. Explain the steps followed.
- Start a new VM with the `Cloud` and `external` interfaces attached. Explain the steps followed.

□

OpenNebula Networks

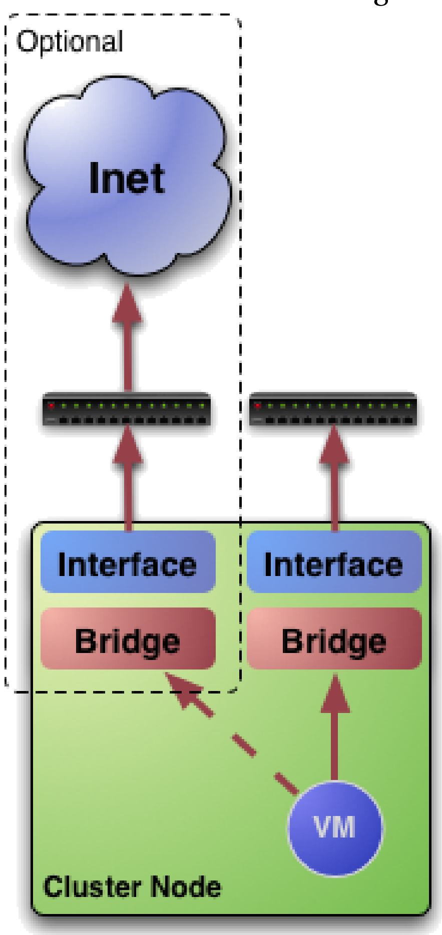
A Virtual Network, or Vnet, in OpenNebula:

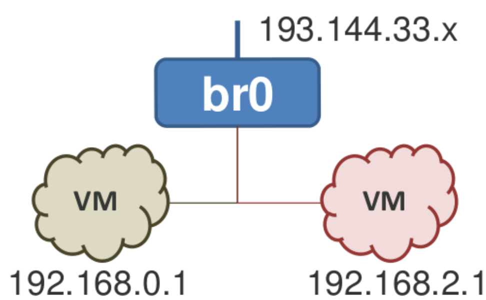
- is used to define a MAC/IP address range that can be used by the VMs.
- A Vnet is associated to a physical network through a bridge.
- Vnets can be isolated (at *layer 2* level)

How to define a Virtual Network on Sunstone:

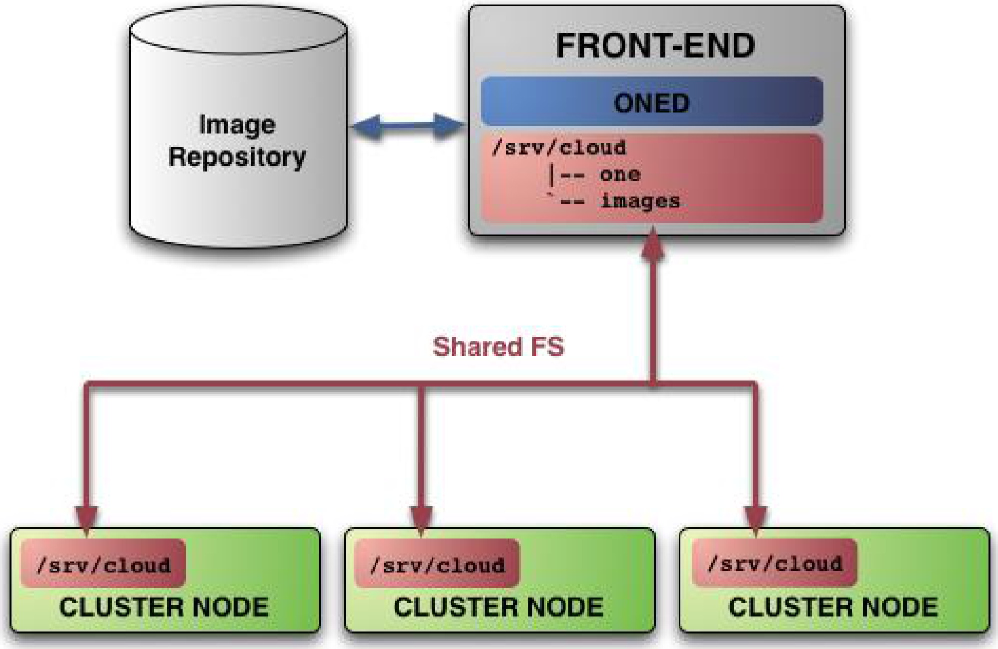
- **Name** of the network
- **Type**
  - **Fixed:** a set of IP/MAC *leases*
  - **Ranged:** to define an address range
- **Bridge:** Name of the physical bridge in the physical host where the VM must connect its network interface.

The Virtual networks can be managed also via the CLI: `onevnet`





OpenNebula Image Repository



Images in the OpenNebula repository

- A VM hard disk can be used as a device to store either an operating system or data.
- Images can be **persistent** and/or public
- The modifications performed to an image can be saved as a new image.

Image types:

- **OS** : Contains a working operating system
- **CDROM** : Read-only data
- **DATABLOCK** : Data storage (hard drive empty or with already existing data)
- **RAMDISK** : File to use as a ramdisk
- **KERNEL** : File to use as a kernel
- **CONTEXT** : File to use in a CDROM drive as a contextualisation file

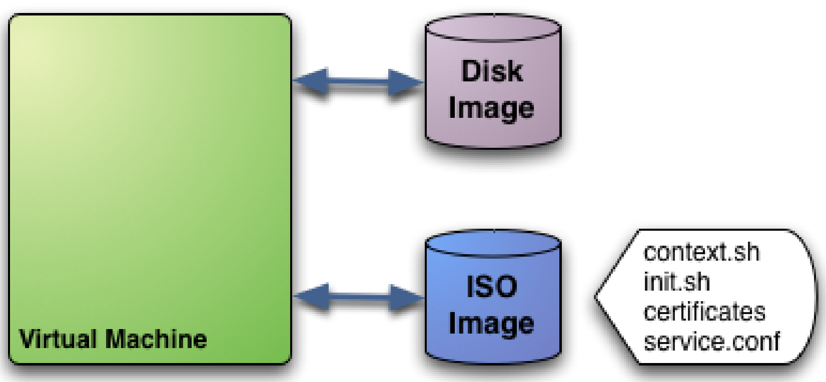
Templates

- Templates are models to define VMs
- The same template can be used to create multiple VMs

- Templates comprise:
  - **Capacity** : Memory and CPU
  - **NICs** (Network interfaces) attached to one or several vLANs
  - A set of **hard drives**
  - An optional file to store a running VM's memory status plus information about the hypervisor.

Contextualisation

- Gives configuration information to a VM that has just been created.
- Examples: boot scripts, connexion ids, network configuration, etc.



3 Create a new Template using Sunstone...

- ... that boots from an existing hard disk that has already an operating system installed on it.
- with two network interfaces.
- with an additional disk in ext3 format.
- Once the VM has started make sure that you can access (`mount`) the additional hard disk.
- Using iptables, make sure the VMs can access the internet through the frontend (Virtualbox OpenNebula VM).

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